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DEFENCE PROPOSAL

Secure Internal chat - SentiZK

Present By:

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INTRODUCTION

1. Internal messaging is widely used in organizations.
2. Current systems use E2EE and MFA but still have security gaps.
3. SentriZK integrates ZKP authentication + AI anomaly detection for enhanced security.

Security Limitations of Current Systems

- Centralized servers store hashed credentials → vulnerable to breaches.
- MFA alone does not fully prevent unauthorized access.

Lack of Embedded Threat Detection

- Anomaly detection is mostly applied to networks/IoT, not chat platforms.
- Malicious activity can move inside normal chat flows undetected.

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Problem Statement

1. Authentication Vulnerability

- ✓ Internal chat platforms rely on centralized servers storing credentials.
- ✓ Breaches like 2023 Slack attack show stolen tokens can expose communications.
- ✓ ZKP authentication can verify users without storing or sending passwords.

2. Lack of Intelligent Threat Detection

- ✓ E2EE protects messages but prevents servers from analyzing threats.
- ✓ Malicious activity (e.g., 2025 Discord malware attack) can go undetected.
- ✓ On-device AI anomaly detection can identify risky behavior locally.

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Significance Of this Project

Enhance Security

- Protects sensitive organizational communications from breaches.
- Eliminates reliance on stored credentials using ZKP authentication.

Proactive Threat Detection

- Detects malicious or suspicious activity without breaking encryption.
- On-device AI anomaly detection improves real-time security.

Contribution to Cybersecurity

- Strengthens system and application security in internal chat platforms.
- Offers insights for SMEs to secure internal communications efficiently.

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Research Objectives

01 To Study

Existing authentication & anomaly detection methods.

Security gaps in internal messaging systems

02 To Develop

Secure chat app with ZKP authentication

On-device AI anomaly detection

03 To Test

Effectiveness against unauthorized access

Detection of suspicious behavior without breaking privacy

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Literature Review
Key Studies

Study	Technique / Model	Key Contribution	Relevance to My FYP
Bhattacharya et al. (2024)	Zero-Knowledge Proofs (zk-SNARKs, zk-STARKs)	ZKP-based authentication without revealing user data	Supports my ZKP login authentication mechanism
Gaurav et al. (2024)	TinyBERT + Aquila Optimization on Android	High-accuracy (96.81%) mobile smishing detection	Shows AI can run efficiently on client/mobile supports my on-device anomaly detection
Natha et al. (2022)	Review of ML/DL anomaly detection	Summarizes models, datasets, challenges (2019 - 2021)	Provides foundation for designing anomaly detection module inside chat system
Alatawi (2022)	Analysis of E2EE & authentication ceremonies	Shows current E2EE apps still weak against MitM attacks	Justifies improving E2EE & authentication with ZKP in my system

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Literature Review Research Gap

Observations from Key Studies Online Behavior

1. **Bhattacharya et al. (2024)** – ZKP authentication exists but not combined with on-device anomaly detection.
2. **Gaurav et al. (2024)** – AI smishing works on mobile but not integrated into secure chat with E2EE.
3. **Natha et al. (2022)** – ML/DL anomaly detection reviewed; most solutions server-based or resource-heavy, not client-friendly.
4. **Alatawi (2022)** – E2EE apps vulnerable to MitM; current authentication not leveraging ZKP.

Identified Gap

1. No system integrates ZKP-based authentication + client-side anomaly detection + E2EE messaging.
2. Existing solutions either lack real-time mobile implementation or secure authentication integration.
3. Research need: Lightweight, mobile-friendly anomaly detection integrated with secure ZKP login and E2EE messaging.

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USE OF SECURITY TOOLS

01 Install protective software

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02 Protect data with encryption

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MONITORING AND RESPONSE PLAN

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01 **Act quickly when a problem is found**

02 **Watch systems regularly for unusual activity**

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CONCLUSION AND NEXT STEPS

Start today

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Take small steps

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Watch for threats

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Protect data together

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THANK YOU
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